



III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	A	D	B	D	B	B	A	C	D	B	C	B	C	C

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
B	A	D	D	C	A	C	A	D	B	A	D	C	B	D

31	32	33	34	35	36	37								
C	A	D	C	B	B	D								

- 1 C Minima is the result of destructive interference between the waves. The phase difference between the waves is π radian or odd multiples of π radian.

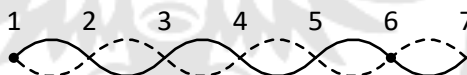
- 2 A Rayleigh criterion

$$\theta = \frac{1.22\lambda}{b}$$

To increase resolving power, θ should be as small as possible (such that the two images can still be resolved even if their angular separation is small. This is achieved with shorter wavelength or larger aperture size b .

Note: The factor of 1.22 is for circular aperture, e.g., our pupil. However, this factor is not important to solve this problem.

- 3 D

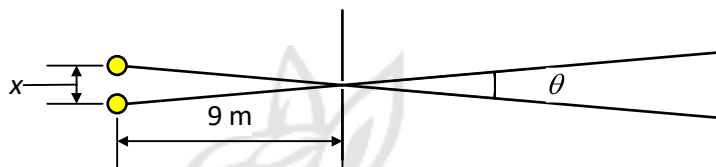


Distance between first and sixth node is $2.5\lambda = 7.5$ m, so $\lambda = 3.0$ m.

Speed of waves

$$v = f\lambda = 300 \times 3.0 = 900 \text{ m s}^{-1}$$

4 B



Using Rayleigh criterion for a single slit, the minimum angle for image to be just resolved,

$$\theta = \frac{\lambda}{b} = \frac{500 \times 10^{-9}}{25 \times 10^{-3}} = 2.0 \times 10^{-5} \text{ rad}$$

Separation

$$x = r\theta = 9.0 \times 2.0 \times 10^{-5} = 0.0018 \text{ m} = 0.18 \text{ mm}$$

5 D Intensity $I \propto A^2$.

$$\text{Maximum intensity } I_0 \propto (2A + A)^2 = 9A^2 \quad I_0 \propto (2A + A)^2 = 9A^2$$

$$\text{Minimum intensity } I_{\min} \propto (2A - A)^2 = A^2 = \frac{1}{9} I_0$$

6 B Rayleigh criterion for a single slit

$$\theta_{\min} = \frac{\lambda}{b} = \frac{500 \times 10^{-9}}{200 \times 10^{-3}} = 2.5 \times 10^{-6} \text{ rad}$$

Angular separation of panels

$$\theta = \frac{s}{r} = \frac{25}{335 \times 10^3} = 7.5 \times 10^{-5} \text{ rad}$$

Since $\theta > \theta_{\min}$, the panels can be resolved.

7 B Stationary waves are formed between the two loudspeakers.

Internodal distance

$$d = \frac{1}{2} \lambda = \frac{v}{2f} = \frac{330}{2 \times 1100} = 0.15 \text{ m}$$

Frequency of detecting loud sound

$$f = \frac{v_{\text{detector}}}{d} = \frac{30}{0.15} = 200 \text{ Hz}$$

8 A Fringe separation

$$x = \frac{\lambda D}{a} \propto \frac{\lambda}{a}$$

So new fringe separation

$$x' = \frac{\lambda'}{\lambda} \cdot \frac{a}{a'} \cdot x = \frac{400}{600} \times \frac{1}{2} \times 4.0 = 1.3 \text{ mm}$$

- 9 C Assuming no end correction for closed tube, resonance occurs when

$$f = (2n-1) \cdot \frac{v}{4L}$$

Length of tube

$$L = (2n-1) \cdot \frac{v}{4f} = (2n-1) \cdot \frac{330}{4 \times 500} = 0.165(2n-1)$$

When $n=1$, $L=0.165$ m

When $n=2$, $L=0.495$ m

When $n=3$, $L=0.825$ m

When $n=4$, $L=1.155$ m

- 10 D Since sources are π out of phase, we get constructive interference when $\Delta = (n + \frac{1}{2})\lambda$ and destructive interference when $\Delta = n\lambda$.

$$|XP - YP| = 1.5 \text{ cm} = 1.5\lambda \Rightarrow \text{constructive interference}$$

$$|XQ - YQ| = 0.0 \text{ cm} \Rightarrow \text{destructive interference}$$

$$|XR - YR| = 1.5 \text{ cm} = 1.5\lambda \Rightarrow \text{constructive interference}$$

$$|XS - YS| = 2.0 \text{ cm} = 2\lambda \Rightarrow \text{destructive interference}$$

- 11 B Assuming no end correction, the lengths of air column when resonance occurs are in the ratio of 1:3:5:7:9. The two lengths of 18 cm and 30 cm correspond to the ratio of 3:5. So the lengths are 6 cm, 18 cm, 30 cm, 42 cm, 54 cm, ...
So, 24 cm does not result in resonance.

- 12 C Since $(n+1)^{\text{th}}$ -order of 400 nm coincides with n^{th} -order 600 nm,

$$(n+1)\lambda_1 = n\lambda_2$$

$$\Rightarrow \frac{n+1}{n} = \frac{\lambda_2}{\lambda_1} = \frac{600}{400} = \frac{3}{2}$$

$$\Rightarrow n = 2$$

Slit separation

$$d = \frac{n\lambda_2}{\sin\theta} = \frac{2 \times 600 \times 10^{-9}}{\sin 30^\circ} = 2.4 \times 10^{-6} \text{ m}$$

- 13 B Since the maxima and minima are regularly spaced,

$$x = \frac{\lambda D}{a}$$

Wavelength

$$\lambda = \frac{xa}{D} = \frac{90 \times 200}{3000} = 6.0 \text{ m}$$

- 14 C From $d \sin\theta = n\lambda$, we can deduce that 1st-order red fringe will coincide with 2nd-order violet fringe. Also the spectrum should range from violet (smaller angular position) to red (larger angular position).
Note: If wavelength of violet is 400 nm instead, then 2nd-order violet fringe will not coincide exactly with 1st-order red fringe.

- 15 C** To be able to observe interference fringes, interfering light waves must be coherent (constant phase difference). However, light waves emitted by two sources are not coherent as light waves are emitted with random phases.

- 16 B** X and Y are adjacent nodes (pressure antinodes). Since internodal distance is $\frac{1}{2}\lambda = 0.33 \text{ m}$, $\lambda = 0.66 \text{ m}$.

Period of signal

$$T = \frac{1}{f} = \frac{\lambda}{v} = \frac{0.66}{330} = 0.0020 \text{ s} = 2.0 \text{ ms} \text{ (4 squares on c.r.o. screen)}$$

- 17 A** Fundamental mode of vibrating string (nodes at both ends)

$$f_s = \frac{v}{2L}$$

Fundamental mode of stationary sound wave in closed-pipe

$$f_c = \frac{v}{4L_c}$$

Since $f_s = f_c$,

$$\frac{v}{2L} = \frac{v}{4L_c}$$

$$\Rightarrow L_c = 0.5L$$

Note: Speed of wave on string is equal to speed of sound.

- 18 D** Fringe separation

$$Z = \frac{\lambda D}{a}$$

To observe the same fringe separation with the same double-slit set up,

$$\frac{D_2}{D_1} = \frac{\lambda_1}{\lambda_2}$$

$$\Rightarrow D_2 = \frac{\lambda_1}{\lambda_2} \times D_1 = \frac{700}{420} \times 1.20 = 2.00 \text{ m}$$

- 19 D** Fringe separation

$$x = \frac{\lambda D}{a} = \frac{600 \times 10^{-9} \times 2.5}{0.40 \times 10^{-3}} = 3.75 \times 10^{-3} \text{ m}$$

$$\Rightarrow y = 5x = 5 \times 3.75 \times 10^{-3} = 1.9 \times 10^{-2} \text{ m}$$

- 20 C** Particles A and B are at adjacent segments of a stationary wave so their phase difference is 180° . Maximum K.E. is directly proportional to its amplitude. Since amplitude of A < amplitude of B, $K_A < K_B$.

- 21 A** To obtain the same aperture size to wavelength ratio as diagram 2, a shorter wavelength in diagram 1 is required. This is achieved by increasing the frequency of vibration of the bar.

- 22 C** Fundamental frequency for closed pipe

$$f_c = \frac{v}{4L}$$

Fundamental frequency for open pipe

$$f_o = \frac{v}{2L} = 2f_c$$

- 23 A** Since fringe separation $x \propto \lambda/a$ (slit-screen distance D is constant),

$$x_2 = \frac{\lambda_2}{\lambda_1} \times \frac{a_1}{a_2} \times x_1 = 0.5 \times 0.5 \times 3.0 = 0.75 \text{ mm}$$

- 24 D** Using $d \sin \theta_n = n\lambda$ and $d = 1/N$, and for the same wavelength λ ,

$$n_1 N_1 = n_2 N_2$$

$$\Rightarrow \frac{N_1}{N_2} = \frac{n_2}{n_1} = 1.5$$

- 25 B** Fundamental frequency of open tube

$$f_o = \frac{v}{2L}$$

Fundamental frequency of closed tube

$$f_c = \frac{v}{4L} = \frac{1}{2} f_o = 128 \text{ Hz}$$

Since only odd harmonics are present in closed tubes, the two lowest frequencies are 128 Hz and 384 Hz.

- 26 A** The highest order diffracted light for longest wavelength

$$n \leq \frac{d}{\lambda} = \frac{1}{N\lambda} = \frac{1}{500000 \times 600 \times 10^{-9}} = 3.3$$

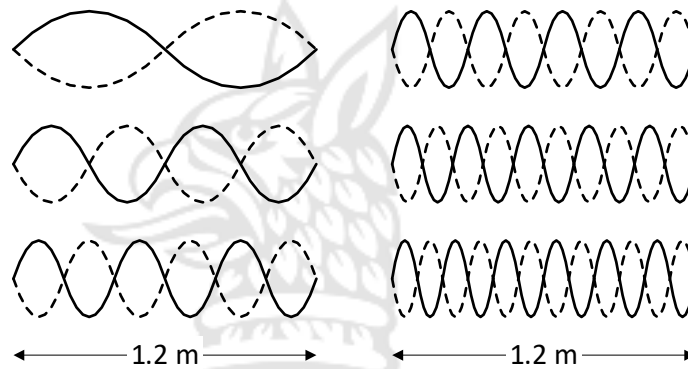
The highest order of complete spectrum is 3rd. So there are 6 complete spectra observed.

Note: Zeroth order maximum is not a spectrum.

- 27 D** To obtain destructive interference at Q, the path difference from P to Q

$$\Delta = (L_1 + L_3) - (L_2 + L_4) = \frac{1}{2}(2m+1)\lambda$$

- 28 C** The figure shows possible modes of vibration where the two ends and the mid-point are nodes.



Since the internodal distance is $\frac{1}{2}\lambda$, the possible wavelengths are 1.2 m, 0.60 m, 0.40 m, 0.30 m, 0.24 m, 0.20 m, ...

- 29 B** To obtain the same aperture size to wavelength ratio as diagram 1, a longer wavelength in diagram 2 is required. This is achieved by decreasing the frequency of vibration of the bar.

- 30 D** For a point source,

$$A \propto \frac{1}{r}$$

So

$$A_2 = \frac{r_1}{r_2} \times A_1 = \frac{r}{2r} \times 8.0 = 4.0 \mu\text{m}$$

- 31 C** The path difference varies from zero (at X or Y) to 6.5λ (centre of rail). At path difference of $n\lambda$, both sources set up a compression or rarefaction at the same time. Hence, the changes in pressure are the largest and sound intensity is a maxima.

Conversely, at path difference of $(n + \frac{1}{2})\lambda$, sound intensity is a minima, i.e., at path difference of 0.5λ , 1.5λ , 2.5λ , 3.5λ , 4.5λ , 5.5λ and 6.5λ . The total number of sound intensity minima is 13.

- 32 A** Using $d \sin \theta = n\lambda$,

$$d = \frac{2\lambda}{\sin \theta_2} = \frac{2 \times 570 \times 10^{-9}}{\sin(69.6/2)} = 2.0 \times 10^{-6} \text{ m}$$

Number of lines per millimetre

$$N = \frac{1 \times 10^{-3}}{d} = \frac{1 \times 10^{-3}}{2.0 \times 10^{-6}} = 500$$

- 33 D** Points between adjacent segments are oscillating π rad out of phase. This means that all points between alternate segments are oscillating in phase.

- 34 C** Since internodal distance is $\frac{1}{2}\lambda$,

$$L = n \times \frac{1}{2}\lambda \quad \Rightarrow \quad \lambda = \frac{2L}{n}$$

The possible wavelengths are 2.0 m, 1.0 m, 0.67 m, 0.50 m, 0.40 m, ...

So, 1.3 m cannot be the wavelength.

- 35 B** Rayleigh's criterion (assuming rectangular aperture)

$$\theta_{\min} = \frac{\lambda}{b}$$

Minimum separation between objects

$$x_{\min} = h\theta_{\min} = \frac{h\lambda}{b} = \frac{120 \times 10^3 \times 5.5 \times 10^{-7}}{0.025} = 2.6 \text{ m}$$

- 36 B** 1.26 m is the minimum distance between the tail lights that can be resolved.

$$\theta_{\min} = \frac{d_{\min}}{r} = \frac{1.26}{14.4 \times 10^3} = 8.75 \times 10^{-5} \text{ rad}$$

- 37 D** Fringe separation

$$x = \frac{\lambda D}{a}$$

The following modifications increases fringe separation.

- Increasing wavelength or decreasing frequency of light.
- Increasing distance between slit and screen.
- Decreasing slit separation.

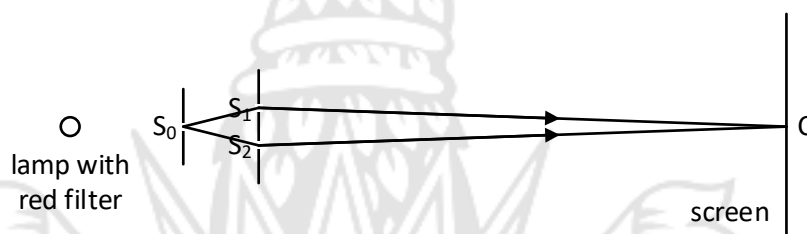
Structured Questions

1. [ACJC/H2/Prelims 2018/P3/7 Modified]

- a. Phase difference between two waves at a point is a measure of the fraction of a complete cycle of one wave being ahead of the other wave.
- b. i. Lamp is not a point source and hence it is not coherent. Incoherent light source cannot produce observable interference pattern.

The single slit acts as a point source with light, and upon reaching S_1 and S_2 , create two coherent light sources.

ii.



iii. Path difference $\Delta = 0$ m

Note: Indicate the units.

iv. 1. C is a point of constructive interference. So we can deduce that the amplitude of unpolarized light due to either S_1 or S_2 is $\frac{1}{2}A$.

With polariser in front of S_2 , light of lower intensity from S_2 reaches C .

So resultant amplitude at C will be less than A .

Note: We cannot compare amplitude of polarised light with that of unpolarised light.

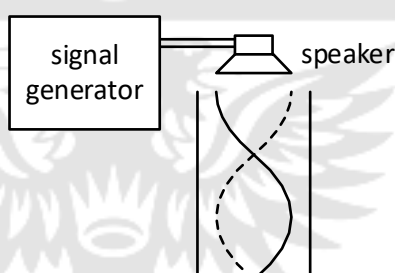
2. Blue filter transmits blue light which has a shorter wavelength than red light. Hence fringe separation is smaller.

Contrast between bright and dark fringes remains the same.

c. i. A stationary wave is the result of interference between two progressive waves of the same type, frequency, amplitude and speed, travelling along the same line but in opposite directions.

ii. 1. Since the waveform corresponds to the position where the microphone picks up the largest pressure variation, the microphone is at a pressure antinode (displacement node).

Assuming no end correction for closed tube, $L = \frac{3}{4}\lambda$. So the stationary wave form is as shown.

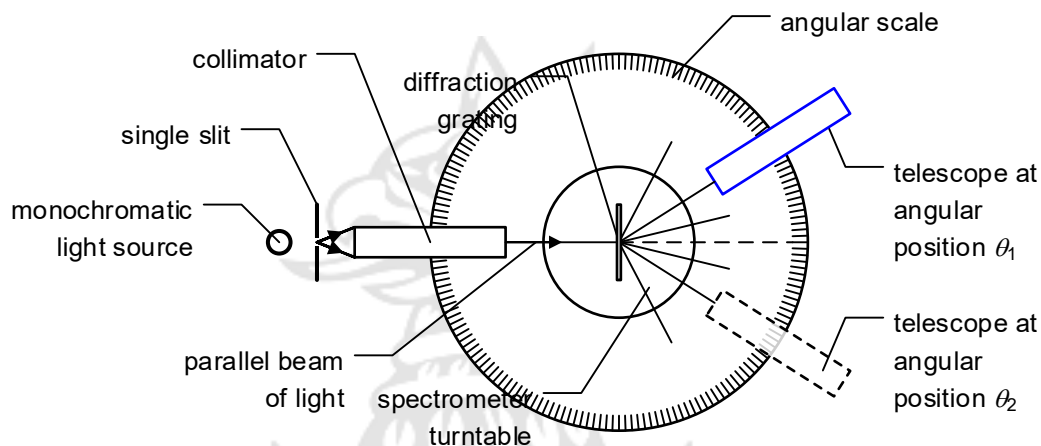


Therefore the microphone must be placed either 30 cm from the open end or at the end of the tube.

2. For a closed tube, length should be odd multiples of $\frac{1}{4}\lambda$ in order to form stationary waves.

For $\lambda = 330$ cm, no stationary waves will be formed in the closed tube of 90 cm.

d.



Measure the angular positions of first (or second) order bright fringe on either side of principal axis.

Use $d \sin \theta = d \sin \left[\frac{1}{2} (\theta_2 - \theta_1) \right] = n \lambda$ to determine wavelength λ , where n is the order of bright fringe, $d = 1/N$ is the slit separation (N is number of slits per unit length, e.g., 600 lines per mm).

2. [AJC/H2/Prelims 2018/P3/5]

- a. The principle of superposition states that when two or more waves of the same kind meet at a point in space, the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point.
- b. i. Since light waves originate from the single slit (in phase) and the path length S_0S_1Y is equal to S_0S_2Y (or the path difference of the light waves arriving at Y is zero), constructive interference occurs at Y .

ii. Fringe separation

$$x = \frac{\lambda D}{a}$$

Wavelength of light

$$\lambda = \frac{xa}{D} = \frac{0.0020 \times 0.0060}{1.8} = 6.7 \times 10^{-7} \text{ m}$$

Note: Fringe separation is 2.0 mm.

- iii. 1. No change to fringe separation.
Intensities of bright fringes decrease while dark fringes remain dark.
2. Since S_0 is closer to S_2 and further from S_1 , the obtain the zeroth order bright fringe where $S_0S_1Y = S_0S_2Y$
 Y must be shifted upwards away from principal axis.
Fringe separation and contrast between bright and dark fringes will remain the same.

3. [CJC/H2/Prelims 2018/P2/4]**a.** Possible answers:

1. The waves or sources must be coherent (constant phase difference)
2. The waves must have similar amplitude (for higher contrast)
3. The waves must overlap and be of the same type (to produce regions of constructive and destructive interference)
4. For transverse waves, they must be unpolarised or polarised in the same plane. (This point applies to EM waves)

b. i. 1. They are coherent since they have a constant phase difference of π radians.2. Since the phase difference of the two waves arriving at P is π radians, there is destructive interference and hence minimum is obtained.**ii.** Path difference $\Delta = QY - PY = \left(\sqrt{6^2 + 3^2} - \sqrt{6^2 + 1^2} \right) \times d = 0.625d$ **iii.** Since waves arrive at Y destructively, and there are 2 maxima before reaching Y, path difference of the waves arriving at Y

$$\Delta = 2.5\lambda = 0.625d$$

Frequency of microwaves

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{0.625d/2.5} = \frac{1.2 \times 10^9}{d} \text{ Hz}$$

4. [CJC/H2/Prelims 2018/P2/5]**a.** Angular position of 1st-order bright fringe

$$\theta_1 = \tan^{-1} \frac{0.46}{2.00} = 13^\circ$$

Wavelength

$$\lambda = d \sin \theta = \frac{\sin 13^\circ}{300 \times 10^3} = 7.5 \times 10^{-7} \text{ m}$$

b. The angular position of 2nd-order bright fringe is greater which results in a lower percentage uncertainty in the angular position.**5. [CJC/H2/Prelims 2018/P3/4]****a.** The current-carrying wire experiences an alternating magnetic force in the vertical plane (by Fleming's left-hand rule) due to alternating current.

This causes the wire to vibrate vertically. This vibration sends progressive waves of the same frequency as the alternating current towards X and Y.

Reflected waves, with the same frequency, speed and amplitude but travelling in opposite direction, superpose to produce stationary wave provided the frequency of the alternating current matches the resonance frequency of the wire.

b. Wavelength $\lambda = 2 \times \text{internodal distance} = 2 \times 0.60 = 1.2 \text{ m}$ **c.** Speed of wave $v = f\lambda = 10 \times 1.2 = 12 \text{ m s}^{-1}$ **d.**This is the 3rd harmonics since $f_3 = 3f_1$.

8. [EJC/H2/Prelims 2018/P3/7* Modified]

- a. i. An electric current that periodically reverses its direction in a circuit (with a frequency)
- ii. The wire carries an alternating current that is perpendicular to the magnetic field between the poles of magnets.

The wire experiences an alternating magnetic force in the vertical plane (by Fleming's left-hand rule) due to alternating current.

This causes the wire to vibrate vertically (up and down), with the same frequency as the alternating current.

- iii. 1. Speed of wave

$$v = f\lambda = f \times 2 \times \text{internodal distance} = 120 \times 2 \times 2.0 = 480 \text{ m s}^{-1}$$

2. This vibration sends progressive waves (of same frequency as a.c.) traveling with speed v along the wire towards X and Y.

Reflected waves, with the same frequency, speed and amplitude but travelling in opposite direction, superpose to produce stationary wave provided the frequency of the alternating current matches the resonance frequency of the wire.

- iv. Wavelength of the waves is the same for the same fundamental mode of vibration. So

$$v = f\lambda \propto f$$

Tension in wire is equal to weight of load

$$T = mg \propto m$$

So if $v \propto \sqrt{T}$, then we need to show $f \propto \sqrt{m}$ or $\frac{f_1}{f_2} = \sqrt{\frac{m_1}{m_2}}$.

$$\frac{f_1}{f_2} = \frac{170}{208} = 0.817$$

$$\sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{2}{3}} = 0.816$$

Since $\frac{f_1}{f_2} \approx \sqrt{\frac{m_1}{m_2}}$, the relationship is valid.

- b. i. Diffraction is the bending (or spreading) of a wave after passing through an aperture or around an obstacle.
- ii. Angular position of 2nd-order bright fringes

$$\theta = \frac{1}{2}(\theta_2 - \theta_1) = \frac{1}{2}(188 - 160) = 14^\circ$$

Slit separation

$$d = \frac{2\lambda}{\sin \theta} = \frac{2 \times 633 \times 10^{-9}}{\sin 14^\circ} = 5.23 \times 10^{-6} \text{ m}$$

Number of lines per metre

$$N = \frac{1}{d} = \frac{1}{5.23 \times 10^{-6}} = 1.91 \times 10^5 \text{ m}^{-1}$$

- iii. The bright fringes are due to the constructive interference from light passing through multiple slits instead of just two slits. Hence the bright fringes produced by diffraction grating is much brighter than that produced by double slits.

9. [HCI/H2/Prelims 2018/P3/3 Modified]

- a. The principle of superposition states that when two or more waves of the same kind meet at a point in space, the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point.

- b. When the path difference is $(n + \frac{1}{2})\lambda$, the waves meet in phase and interfere constructively. The detected intensity is high.

When the path difference is $n\lambda$, the waves meet in antiphase and interfere destructively. The detected intensity is low.

As the reflector moves towards the source/detector, the path difference between the direct and reflected waves decreases and so the intensity at the detector alternates between high and low.

- c. i. Path difference

$$\Delta = 5\sqrt{2} - 5 = n \times 0.59$$

$$\Rightarrow n = 3.5$$

So the intensity at the detector is a maximum.

- ii. Maximum path difference between direct and reflected wave is 5.0 m (when the detector is directly in front of the source).

$$n_{\max} = \frac{\Delta_{\max}}{\lambda} = \frac{5.0}{0.59} = 8.5$$

As the detector approaches the source, the intensity of reception changes from maximum to minimum and back to maximum a total of 5 times.

Note: Maximum occurs when $n = 4.5, 5.5, 6.5, 7.5, 8.5$.

10. [HCI/H2/Prelims 2018/P3/4]

- a. i. Uncertainty in L

$$\Delta L = \frac{1}{2}(L_{\max} - L_{\min}) = \frac{1}{2}(6.2 - 5.8) = 0.2 \text{ m}$$

- ii. Since y is small, angular separation is also small.

$$\theta = \frac{y}{L}$$

$$\Rightarrow \frac{\Delta \theta}{\theta} = \frac{\Delta y}{y} + \frac{\Delta L}{L}$$

Fractional uncertainty in y

$$\frac{\Delta y}{y} = \frac{0.5}{2.0} = 0.25$$

Average value of L

$$L = \frac{1}{5}(6.2 + 5.8 + 6.1 + 5.9 + 6.1) = 6.0 \text{ m}$$

Fractional uncertainty in L

$$\frac{\Delta L}{L} = \frac{0.2}{6.0} = 0.03$$

Since the fractional uncertainty in y is 1 order higher than that in L , it is reasonable to ignore the uncertainty in L .

iii. Minimum angular separation

$$\theta_{\min} = \frac{y}{L_{\max}} = \frac{2.0 \times 10^{-3}}{6.0} = 3.3 \times 10^{-4} \text{ rad}$$

$$\Delta\theta_{\min} = \frac{\Delta y}{y} \times \theta_{\min} = 0.25 \times 3.3 \times 10^{-4} = 0.8 \times 10^{-4} \text{ rad}$$

So $\theta_{\min} = (3.3 \pm 0.8) \times 10^{-4} \text{ rad}$.

Note: You may use min-max method to calculate $\Delta\theta_{\min}$.

Note: You may include the uncertainty in L in your calculation.

- b. i. The two diffraction patterns are just resolved when the central maximum of one pattern coincides with the first minimum of the other pattern.

ii. Minimum angle

$$\theta_{\min} = \frac{\lambda}{b} = \frac{700 \times 10^{-9}}{3 \times 10^{-3}} = 2 \times 10^{-4} \text{ rad}$$

11. [IJC/H2/Prelims 2018/P3/5 Modified]

- a. i. A and B are vibrating in phase while A and C are vibrating π rad out of phase (antiphase).
 ii. Use a high-speed video camera to record the stationary wave. Play back the clip at normal speed which will show A and B moving in phase while A and C moving in opposite direction.
- b. i. The amplitude at all points of a progressive wave is constant.
 The amplitude of any point in a stationary wave varies from zero at the node to twice the amplitude of the constituent progressive wave at the antinode.
- ii. The vibrating pin sends a continuous progressive wave towards the pulley. The wave is reflected by the pulley so a reflected progressive wave of the same frequency and amplitude travels back with the same speed towards the vibrating pin.
 Superposition of the incident wave and reflected wave produces a stationary wave provided the frequency of the vibrating pin matches the natural frequency of the wire.

12. [JJC/H2/Prelims 2018/P2/5 Part]

- b. i. Diffraction is the bending (or spreading) of a wave after passing through an aperture or around an obstacle.
 ii. From Fig. 12.3, vertical distance between central maximum and 1st-order maximum is 0.70 m.

Angular separation

$$\theta = \tan^{-1} \frac{0.70}{2.50} = 15.64^\circ$$

Since

$$d \sin \theta = \frac{1}{N} \sin \theta = n\lambda$$

$$\Rightarrow N = \frac{\sin \theta}{n\lambda} = \frac{\sin 15.64^\circ}{590 \times 10^{-9}} = 457000 \text{ lines m}^{-1}$$

Note: You may use the 2nd or 3rd-order maxima.

- iii. 1. The spacing and angular separation between orders is smaller in the vertical direction (due to grating B).
 Since $d \sin \theta = n\lambda$, this implies that grating B has larger slit separation.
 Note: You may make your deduction based on the maximum order in the vertical and horizontal direction of 3 and 2, respectively.

2. Any one of the following:

- Smaller separation between bright fringes.
- Greater number of bright fringes on the screen.
- Equal separation between orders of bright fringes in both horizontal and vertical direction.
- Similar intensity of bright fringes.

13. [MI/H2/Prelims 2018/P2/4]

a. When resonance occurs (no end correction),

$$D = \left(n + \frac{1}{2}\right) \times \frac{1}{2} \lambda_1 = \left(n + \frac{3}{2}\right) \times \frac{1}{2} \lambda_2$$

where n is the number of internodal distance.

$$\left(n + \frac{1}{2}\right) \times \frac{v}{f_1} = \left(n + \frac{3}{2}\right) \times \frac{v}{f_2}$$

$$\Rightarrow f_2 \left(n + \frac{1}{2}\right) = f_1 \left(n + \frac{3}{2}\right)$$

$$\Rightarrow 90.8 \left(n + \frac{1}{2}\right) = 70.6 \left(n + \frac{3}{2}\right)$$

$$\Rightarrow n = 3$$

Depth of well

$$D = \left(n + \frac{1}{2}\right) \times \frac{1}{2} \lambda_1 = \frac{7}{2} \times \frac{1}{2} \times \frac{343}{70.6} = 8.50 \text{ m}$$

b. i. There is more than one line of constructive interference apart from the centre line between the emitters. The AGV might align itself to these other lines of constructive interference instead.

ii. 1. The intensity of signal X is lower indicating that it has travelled further to reach the AGV. So, signal X is from emitter B.

2. $\pi/3$ rad

3. A phase difference of $\pi/3$ rad corresponds to a path difference

$$\Delta = \frac{\Delta\phi}{2\pi} \times \lambda = \frac{1}{6} \lambda$$

$$\Rightarrow \lambda = 6 \times \left(\sqrt{50^2 + 7.0^2} - \sqrt{50^2 + 5.0^2} \right) = 1.43 \text{ m}$$

Frequency of wave

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{1.43} = 2.10 \times 10^8 \text{ Hz}$$

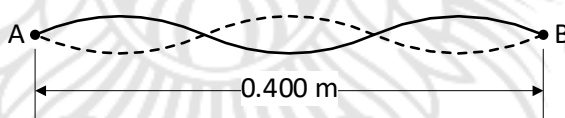
14. [MJC/H2/Prelims 2018/P2/3]

a. Any two of the following:

	Progressive Waves	Stationary Waves
Wave profile	Propagates with velocity of wave.	Does not propagate.
Energy of wave	Transports energy in direction of wave propagation.	Does not transport energy.
Amplitude of oscillation	Every point oscillates with the same amplitude.	Amplitude varies from zero at the nodes to maximum at the antinodes.
Wavelength	All particles within one wavelength have different phases.	All particles between two adjacent nodes have the same phase. Particles in adjacent segments have a phase difference of π rad.
Phase of oscillation	All points vibrate in s.h.m. with the same frequency as the wave.	Except for the nodes which are at rest, all points vibrate in s.h.m. with the same frequency as the progressive waves that give rise to it.
Frequency of oscillation	Equals to the distance between adjacent points which have the same phase.	Equals to twice the distance between a pair of adjacent nodes or antinodes.

c. i. Stationary waves are formed when length AB is an integer multiple of half wavelengths.

ii.



The waveform of the stationary wave oscillating with the third lowest frequency.

Wavelength of (progressive) wave

$$\lambda = \frac{2}{3}L = \frac{2}{3} \times 0.400 = 0.267 \text{ m}$$

iii. Stationary wave is formed by two identical waves (same type, amplitude, frequency, wavelength and/or speed) moving in opposite direction.

The product of frequency and wavelength is the speed of the individual progressive waves that results in the stationary wave.

15. [NJC/H2/Prelims 2018/P3/7 Modified]

- a. i. When the wavelength of the wave is comparable to slit width, diffraction is most significant.
 ii. Low frequency notes have longer wavelengths which are comparable to the width of door gap. These notes experience greater diffraction when passing through the door and can be heard down the corridor although there is no line of sight to the band.
- b. i. Distance D must be much longer than slit separation of the grating.

- ii. Slit separation $d = 1/p$

Path difference

$$\Delta = d \sin \theta = \frac{\sin \theta}{p}$$

- iii. Constructive interference occurs when $\Delta = n\lambda$.

$$\sin \theta = np\lambda$$

where $n = 0, 1, 2, \dots$

- c. i. Angular speed

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{3.0} = 2.1 \text{ rad s}^{-1}$$

- ii. Zeroth-order maximum: peak C. 1st-order maxima: peaks B and D. 2nd-order maxima: peaks A and E. Intensity of maxima decreases with increasing order due to diffraction effect.

- iii. Angular position of 1st-order to central maximum

$$\theta_1 = \omega t_1 = \frac{2}{3}\pi \times 1.7 \times 0.10 = 0.36 \text{ rad}$$

Angular position of 2nd-order to central maximum

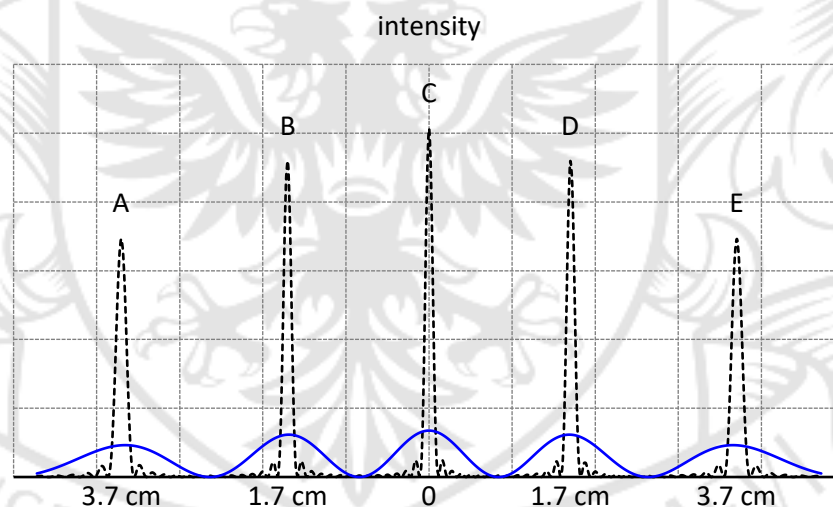
$$\theta_2 = \omega t_2 = \frac{2}{3}\pi \times 3.7 \times 0.10 = 0.77 \text{ rad}$$

- iv. Using the equation in iii.,

$$\lambda = \frac{\sin \theta_2}{2p} = \frac{\sin(0.77)}{2 \times 550000} = 6.3 \times 10^{-7} \text{ m}$$

- v. Peak E is preferred as the angle position (from central maximum) is larger and the percentage uncertainty in calculating the wavelength is smaller.

vi.



Fringe separation remains the same since the slit separation remain the same.

Peaks of lower intensity and broader.

16. [NYJC/H2/Prelims 2018/P2/4]

- a. The oscillations of the electromagnetic fields of polarized light are in a single plane.
- b. The electric field components of the two electromagnetic waves are perpendicular to each other.
The vector sum of these components does not produce maxima and minima because these components are not in the same or in the opposite directions.

- c. i. Wavelength

$$\lambda = \frac{xa}{D} = \frac{0.90 \times 10^{-2} \times 0.10 \times 10^{-3}}{1.5} = 6.0 \times 10^{-7} \text{ m}$$

- ii. The bright fringes are not as bright while the dark fringes are not as dark as complete cancellation of waves to produce dark fringes is no longer possible. So, the contrast of the fringes decreases.

17. [NYJC/H2/Prelims 2018/P3/8 Modified]

- a. i. 1st-order diffraction minimum occurs at 3.0 mm.

Angular position of 1st-order minimum

$$\theta = \tan^{-1} \left(\frac{0.0030}{2.4} \right) = 1.25 \times 10^{-3} \text{ rad}$$

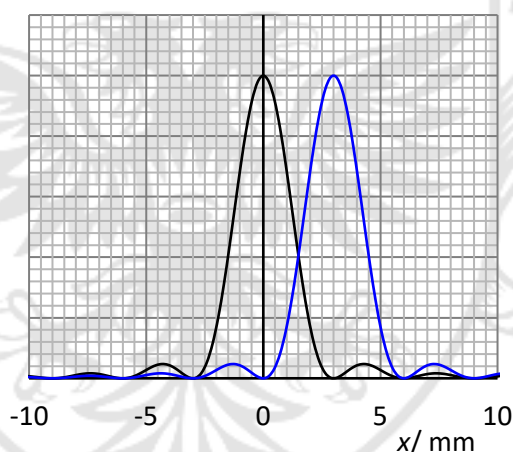
Using $\sin \theta = \frac{\lambda}{b}$, slit width

$$b = \frac{\lambda}{\sin \theta} = \frac{630 \times 10^{-9}}{\sin(1.25 \times 10^{-3})} = 5.0 \times 10^{-4} \text{ m}$$

- ii. 1. Width of central maximum increases since the angular position of 1st-order minimum increases.
Intensity of central maximum decreases as less light passes through a narrower slit.
2. Width of central maximum decreases since the angular position of 1st-order minimum decreases.
Diffraction pattern is violet.

- b. i. The two diffraction patterns are just resolved when the central maximum of one pattern coincides with the first minimum of the other pattern.

ii.



- iii. When the diffraction pattern is just resolved, the angular separation of the sources is equal to the angular separation of the 1st-order minimum from the central maximum.

$$\sin \theta = \frac{\lambda}{b} = \frac{d_{PQ}}{D}$$

$$\Rightarrow b = \frac{\lambda D}{d_{PQ}} = \frac{630 \times 10^{-9} \times 1.5}{5.0 \times 10^{-3}} = 1.9 \times 10^{-4} \text{ m}$$

c. i. Path difference

$$\Delta = \sqrt{1.50^2 + 0.80^2} - 1.50 = 0.20 \text{ m}$$

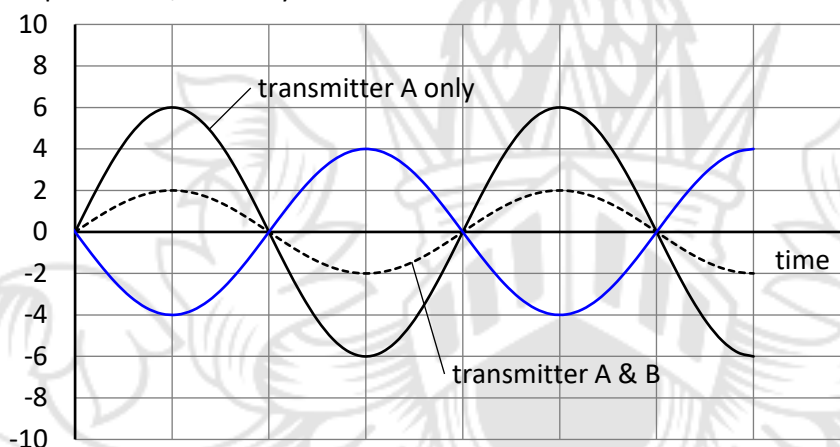
ii. Wavelength of microwave

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{3.00 \times 10^9} = 0.10 \text{ m}$$

That is, the path difference $\Delta = 2\lambda$.

Since no phase difference (or a phase difference of 4π) is introduced due to the longer path from transmitter B to R and the waves arrived at R in antiphase, the waves must be emitted in antiphase.

iii. displacement/ arbitrary units

iv. $\frac{\text{intensity of microwave due to transmitter A}}{\text{intensity of microwave due to transmitter B}}$

$$= \left(\frac{\text{amplitude of microwave due to transmitter A}}{\text{amplitude of microwave due to transmitter B}} \right)^2 = \left(\frac{6}{4} \right)^2 = 2.25$$

18. [PJC/H2/Prelims 2018/P3/4 Modified]

- a. The principle of superposition states that when two or more waves of the same kind meet at a point in space, the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point.

- a. i. Resultant displacement

$$x_R = (-2.6 + 1.8) \times 10^{-4} = -0.8 \times 10^{-4} \text{ cm}$$

Note: $x_R = -0.80 \times 10^{-4} \text{ cm}$ is accepted.

- ii. Since the waves produced displacements that are antiphase, resultant amplitude at P

$$A_R = 1.0 \times 10^{-4} \text{ cm}$$

Since $I \propto A^2$,

$$I_R = I \left(\frac{A_R}{A_A} \right)^2 = I \left(\frac{1}{3} \right)^2 = 0.11I$$

- iii. 1. For 3rd-order minimum, path difference $\Delta = \frac{5}{2} \lambda$.

$$\sqrt{20.0^2 + 31.7^2} - \sqrt{20.0^2 + 28.7^2} = 2.5\lambda$$

$$\Rightarrow \lambda = 1.0 \text{ m}$$

Speed of sound

$$v = f\lambda = \frac{1.0}{3.0 \times 10^{-3}} = 330 \text{ m s}^{-1}$$

2.

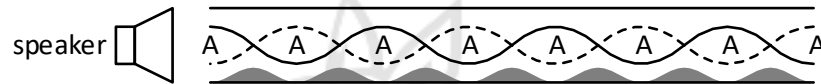


The intensity alternates between minimum at C to maximum to minimum a total of 6 times with the 7th minimum at D.

Note: Stationary wave is formed along CD with C and D being the nodes (minimum).

19. [RVHS/H2/Prelims 2018/P3/7 Modified]

a. i.



ii. From Fig. 19.1, distance between 6 heaps is 5 internodal distance.

$$2.5\lambda = 39.0 \text{ cm}$$

$$\Rightarrow \lambda = 15.6 \text{ cm}$$

iii. Speed of sound

$$v = f\lambda = 2140 \times 0.156 = 334 \text{ m s}^{-1}$$

iv. The length of the tube spans 7 internodal distance, i.e., $L = \frac{7}{2}\lambda = 54.6 \text{ cm}$.When the tube resonates with fundamental frequency, $\lambda = 2L$.

Fundamental frequency of tube

$$f_o = \frac{v}{2L} = \frac{334}{2 \times 0.546} = 306 \text{ Hz}$$

v. Fundamental frequency of closed tube

$$f_c = \frac{v}{4L} = \frac{334}{4 \times 0.546} = 153 \text{ Hz}$$

Resonant frequency is odd multiples of f_c . Since $2140 \text{ Hz} = 14 \times 153 \text{ Hz}$, resonance does not occur and clearly defined heaps of dust are no longer observed.

b. i. $\pi \text{ rad}$

ii. 3 internodal distance spans a distance of 4.20 cm, i.e.,

$$1.5\lambda = 4.20 \text{ cm}$$

$$\Rightarrow \lambda = 2.80 \text{ cm}$$

Speed of microwaves

$$v = f\lambda = 10 \times 10^9 \times 0.280 = 2.80 \times 10^8 \text{ m s}^{-1}$$

iii. A high intensity is registered when the plate is moved by $\frac{1}{2}\lambda = 1.40 \text{ cm}$.

So, frequency of fluctuation

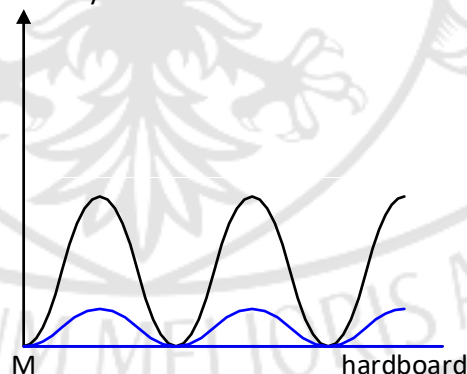
$$f = \frac{v}{\frac{1}{2}\lambda} = \frac{1.00}{1.40} = 0.714 \text{ Hz}$$

iv. Since $I \propto A^2$,

$$\frac{\text{amplitude of detected wave after hardboard insertion}}{\text{amplitude of detected wave before hardboard insertion}} = \sqrt{\frac{1}{4}} = 0.50$$

v.

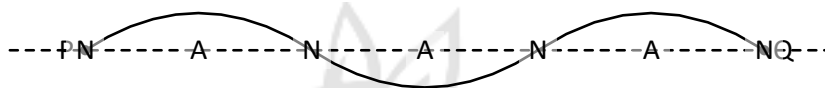
intensity/arbitrary units



Maximum intensity after hardboard insertion is $\frac{1}{4}$ maximum intensity before hardboard insertion.

20. [SRJC/H2/Prelims 2018/P2/4 Modified]

a. i.



ii. Period of oscillation

$$T = \frac{1}{f} = \frac{1}{50.0} = 20 \text{ ms}$$

A quarter of a cycle (5.0 ms) later,



b. i. The reflected sound waves have the same amplitude, wavelength and frequency as the incident sound waves. These waves move in opposite direction and superpose in the space between the speaker and the metal plate. At some positions, these waves meet in phase, resulting in constructive interference and regions of high intensity (antinodes). At other positions (midway between antinodes), these waves meet in antiphase, resulting in destructive interference and regions of low intensity (nodes).

ii. 1. Frequency of sound

$$f = \frac{1}{T} = \frac{1}{0.40 \times 10^{-3}} = 2500 \text{ Hz}$$

2. Distance between antinode and node is $\frac{1}{4}\lambda$.

Speed of sound

$$v = f\lambda = 2500 \times 4 \times 0.033 = 330 \text{ m s}^{-1}$$

21. [SRJC/H2/Prelims 2018/P3/3]

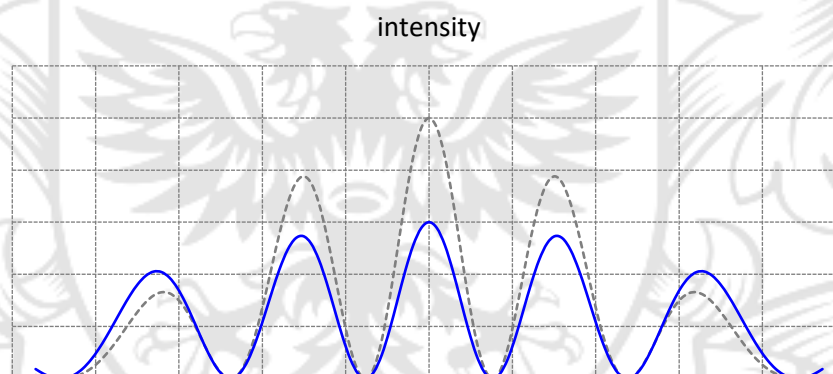
a. i. Path difference

$$\Delta = \frac{1}{2}\lambda = \frac{1}{2} \times 580 \times 10^{-9} = 2.9 \times 10^{-7} \text{ m}$$

ii. Using $x = \lambda D/a$,

$$D = \frac{xa}{\lambda} = \frac{2 \times 2.0 \times 10^{-3} \times 0.41 \times 10^{-3}}{580 \times 10^{-9}} = 2.8 \text{ m}$$

iii.



Lower central maximum; more gradual decrease in intensities of maxima; no change to positions of maxima.

- b. i. Using $d \sin \theta = n\lambda$, the graph of $\sin \theta$ vs. λ is a straight line with gradient of n/d and zero vertical intercept.

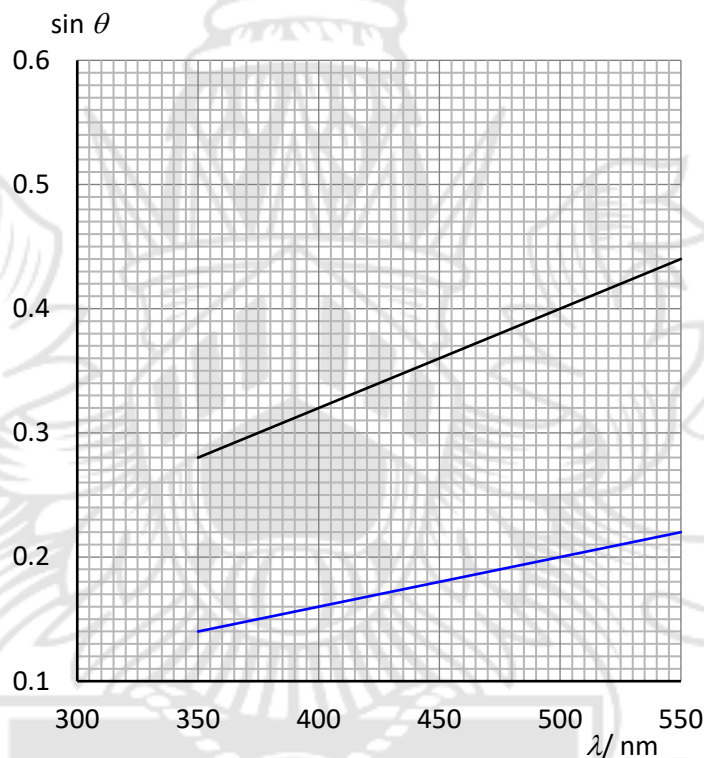
Gradient of graph

$$m = \frac{0.440 - 0.280}{(550 - 350) \times 10^{-9}} = \frac{0.160}{200 \times 10^{-9}} = 8.0 \times 10^5$$

Slit separation

$$d = \frac{n}{\text{gradient}} = \frac{2}{8.0 \times 10^5} = 2.5 \times 10^{-6} \text{ m}$$

ii.

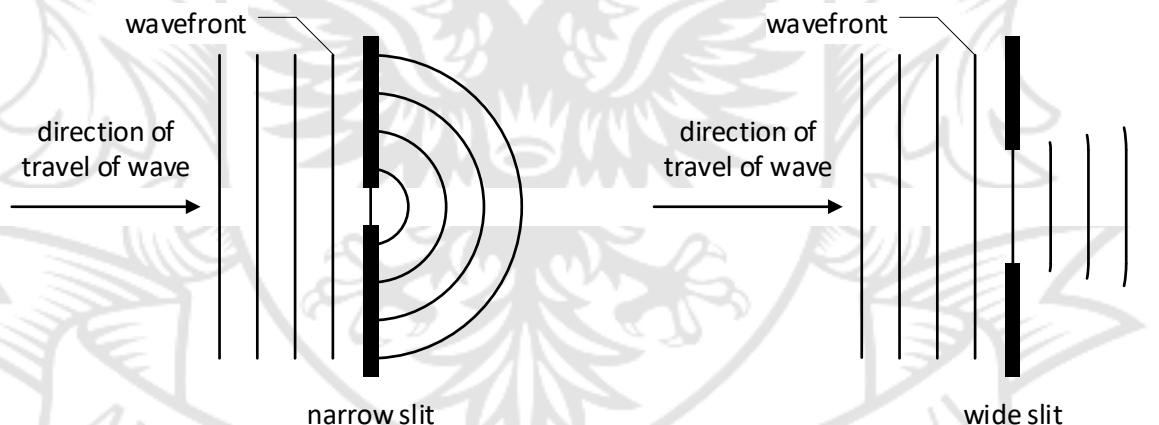


Gradient is halved ($2/d$ vs. $1/d$) with zero vertical intercept, i.e., all plots are halved in values.

22. [TPJC/H2/Prelims 2018/P3/7 Modified]

- a. i. Diffraction is the bending (or spreading) of a wave after passing through an aperture or around an obstacle.

ii.

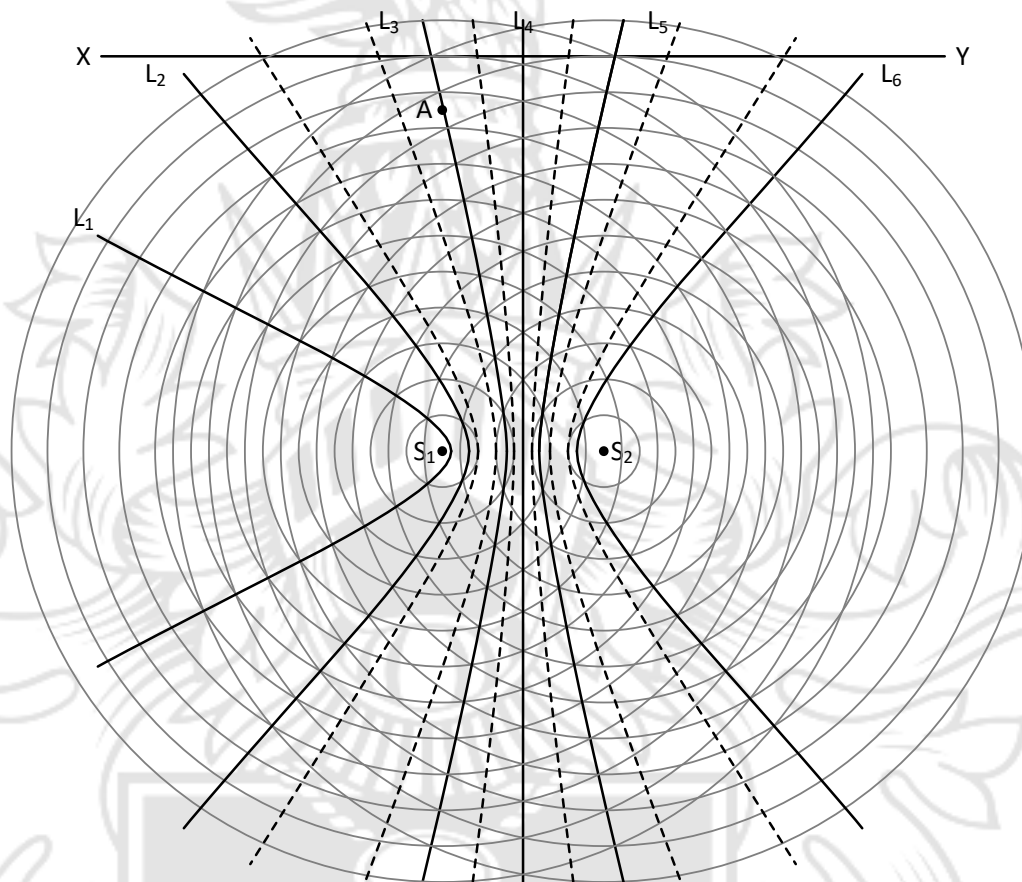


Greater degree of spreading for narrow slit; equal spacing between wavefronts.

- iii. With wide slits, there is little spreading (diffraction) of water waves and hence the wavefronts after passing through the slits do not overlap sufficiently to interfere.

Narrowing the slits increases the spreading of waves such that the wavefronts overlap to produce regions of constructive and destructive interference which leads to observable interference patterns.

- b. i. Coherent sources emit waves with constant phase difference.
The waves have the same frequency, wavelength and speed.
- ii. 1. L_1 to L_6 are antinodal lines joining all points of constructive interference/maxima.
Point A lies on L_3 and is a point of constructive interference.
2. $S_1A - S_2A = \lambda$
- 3.

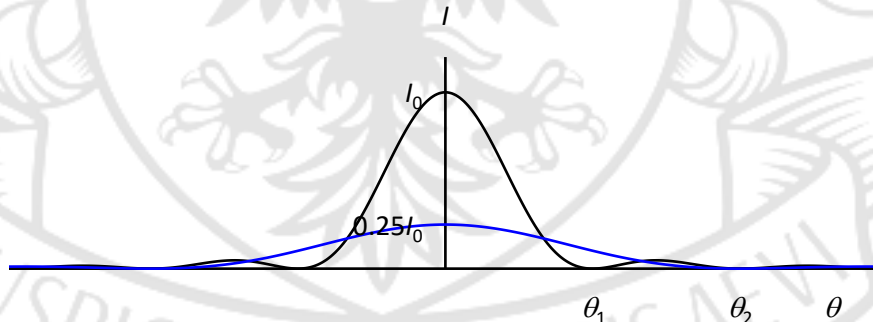


Any of the dotted lines.

Note: L_3 and L_5 correspond to $\Delta = \lambda$. L_2 and L_6 correspond to $\Delta = 2\lambda$. L_1 corresponds to $\Delta = 4\lambda$.

4. Spacing between maxima decreases.

- c. i. Wavelength of light
 $\lambda = b \sin \theta = 0.20 \times 10^{-3} \times \sin 0.17^\circ = 5.9 \times 10^{-7} \text{ m}$
- ii. Width of central fringe
 $w = 2D \tan \theta = 2 \times 0.75 \times \tan 0.17^\circ = 4.5 \times 10^{-3} \text{ m}$
- iii.



Lower maximum intensity ($\frac{1}{4} I_0$); wider centre maximum.

Explanation: Light energy is halved as slit width is halved. This energy is spread over double the area so the maximum intensity is $\frac{1}{4} I_0$.

23. [VJC/H2/Prelims 2018/P3/3 Modified]

- a. Separation between regions of loud sounds

$$x = \frac{\lambda D}{a} = \frac{vD}{fa}$$

Frequency of sound

$$f = \frac{vD}{xa} = \frac{340 \times 20}{1.5 \times 1.2} = 3800 \text{ Hz}$$

- b. $2A$

- c. i. From $I = kA^2$,

$$A_1 = \sqrt{\frac{\frac{1}{2}I}{k}} = \sqrt{\frac{\frac{1}{2}A^2}{k}} = \frac{A}{\sqrt{2}} = 0.71A$$

Note: $P \propto I$ so half power implies half intensity.

- ii. New resultant amplitude

$$A_r = A + 0.71A = 1.71A$$

- iii. Original resultant intensity

$$I = k \times (2A)^2 = 4kA^2$$

New resultant intensity

$$I = k \times (1.71A)^2 = 2.92kA^2 = 0.73I$$

24. [YJC/H2/Prelims 2018/P3/8 Modified]

- a. Speed of sound for each data point:

$$v_1 = \frac{2d_1}{t_1} = \frac{2 \times 10.0}{60.0 \times 10^{-3}} = 333 \text{ m s}^{-1}$$

$$v_2 = \frac{2d_2}{t_2} = \frac{2 \times 20.0}{120.0 \times 10^{-3}} = 333 \text{ m s}^{-1}$$

$$v_3 = \frac{2d_3}{t_3} = \frac{2 \times 30.0}{175.0 \times 10^{-3}} = 343 \text{ m s}^{-1}$$

Average speed of sound

$$\langle v \rangle = \frac{1}{3}(333.3 + 333.3 + 342.9) = 337 \text{ m s}^{-1}$$

Note: You may calculate the speed of sound by using the gradient.

- b. i. Separation between maxima

$$x = \frac{\lambda D}{a}$$

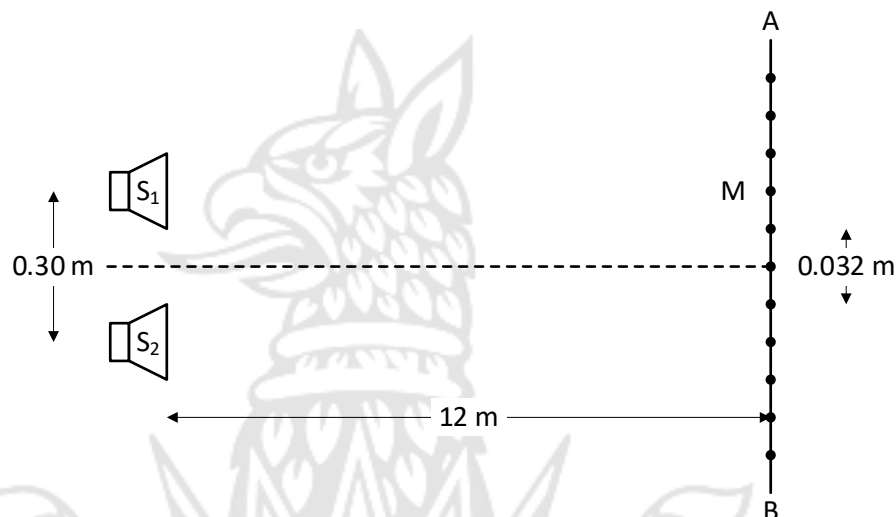
Wavelength of sound

$$\lambda = \frac{xa}{D} = \frac{0.016 \times 0.30}{12} = 0.040 \text{ m}$$

- ii. Speed of sound

$$v = f\lambda = 8300 \times 0.040 = 330 \text{ m s}^{-1}$$

iii. 1.



2. The path difference $S_2M - S_1M$ is integer multiple of wavelengths. Since the sources are emitting waves in phase, the waves arrive in phase at M , resulting in constructive interference and a maximum.

iv. Wavelength of sound

$$\lambda = \frac{v}{f} = \frac{330}{300} = 1.1\text{ m}$$

There is no succession of maxima and minima as the maximum path difference of 0.30 m is less than the wavelength of sound.

c. i. Angular position of 1st-order maximum for 500 nm

$$\theta = \sin^{-1}(N\lambda) = \sin^{-1}(10000 \times 500 \times 10^{-9}) = 0.286^\circ$$

ii. Angular position of 1st-order maximum for 700 nm

$$\theta = \sin^{-1}(N\lambda) = \sin^{-1}(10000 \times 700 \times 10^{-9}) = 0.401^\circ$$

Angular subtended by 1st-order spectrum

$$\Delta\theta = 0.401 - 0.286 = 0.115^\circ$$

iii. Width of 1st-order spectrum on screen

$$x = 2.0 \tan 0.401^\circ - 2.0 \tan 0.286^\circ = 4.0 \times 10^{-3}\text{ m}$$

iv. As the light source is further from the slit, a lower intensity light passes through the slit.

Assuming that light is emitted uniformly in all direction, the new light intensity is

$$I = I_0 \left(\frac{r_0}{r} \right)^2 = I_0 \left(\frac{0.20}{0.50} \right)^2 = 0.16 I_0$$

where I_0 is the original light intensity.